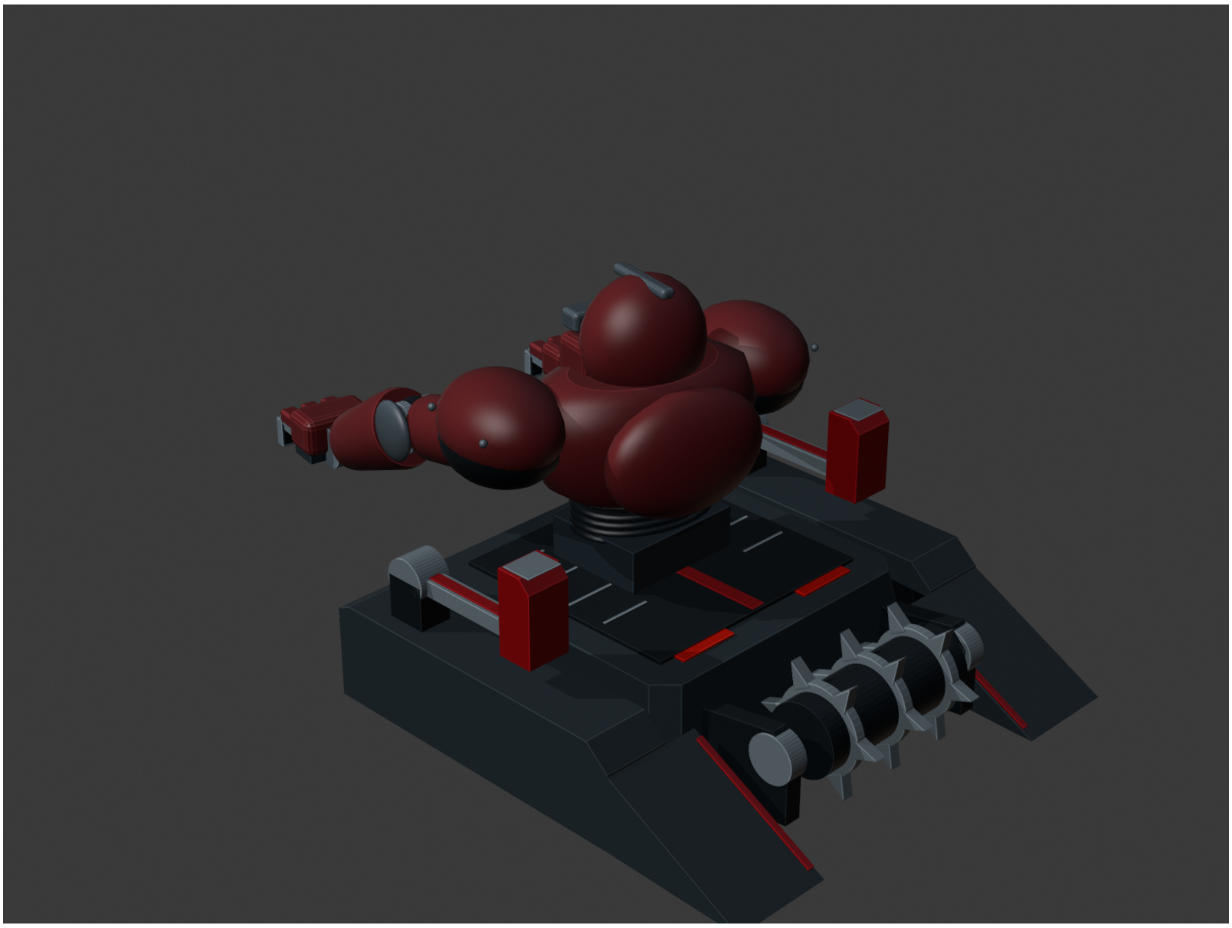
SENTINELT

Robo Wars RC Combat Robot

Project Developer: TG Assegaai



318.16 mm x 495.00 mm footprint | 180/180 frames PASS

Robo Wars Design Constraints

Requirements translated directly into the SentinelT design

- Maximum footprint: 500 mm x 500 mm.
- Mass strictly below 5 kg.
- Human-operated remote control limited to 2.4 GHz.
- Accessible main ON/OFF power isolation.
- Mandatory Emergency Stop / immediate power-cut mechanism.
- No projectiles, flames or liquids.

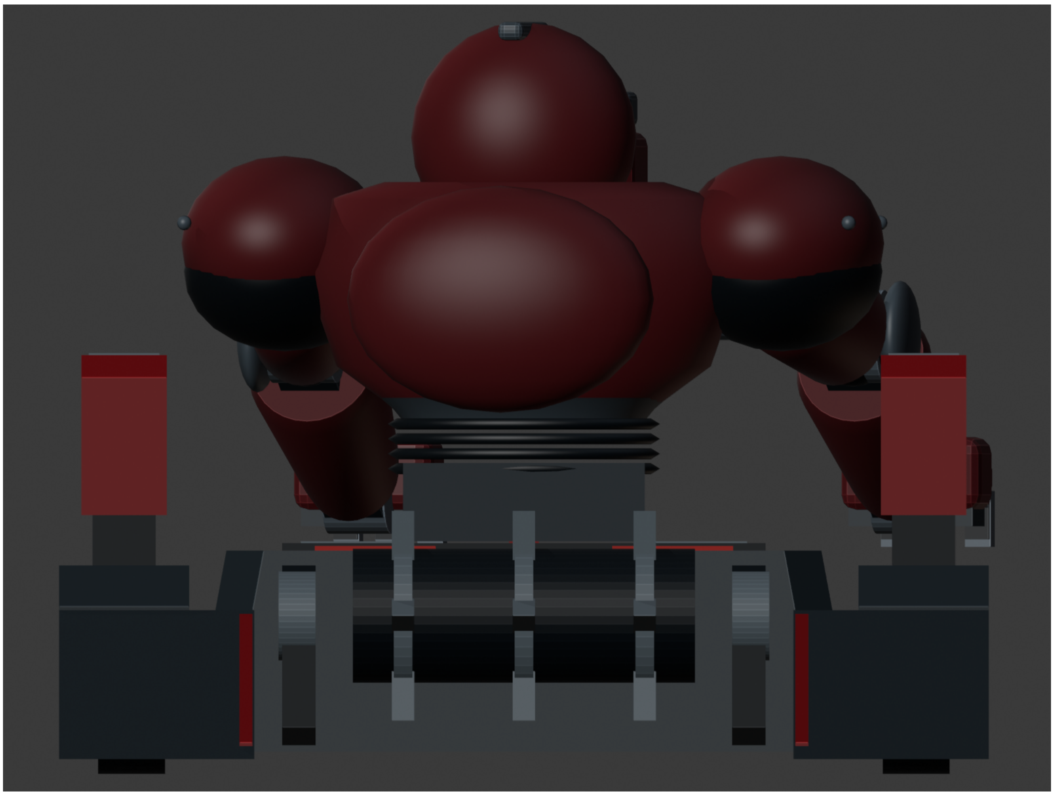
Verified: 318.16 mm x 495.00 mm

Engineering mass budget: 4.650 kg

Physical mass will be verified after hardware implementation.

Solution & Buildability

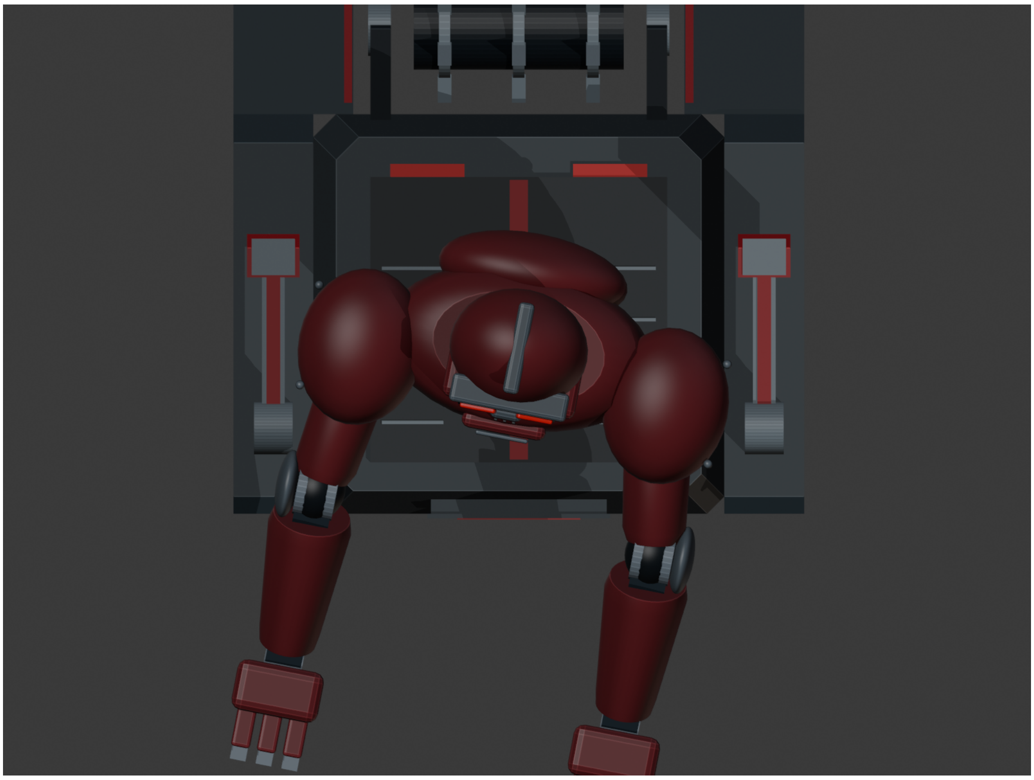
Compact armor, protected drive layout and serviceable packaging



- 6 mm HDPE chassis and armor concept with metal shafts, brackets and fasteners where required.
- Protected drivetrain and sloped guards reduce exposure to direct impacts.
- Bolted construction supports repair, inspection and component replacement.
- Internal service bay packages battery, ESC, receiver, fuse, E-Stop hardware and distribution.
- Guarded active-mechanism provision is integrated into the chassis concept.

Mechanical CAD & Simulation

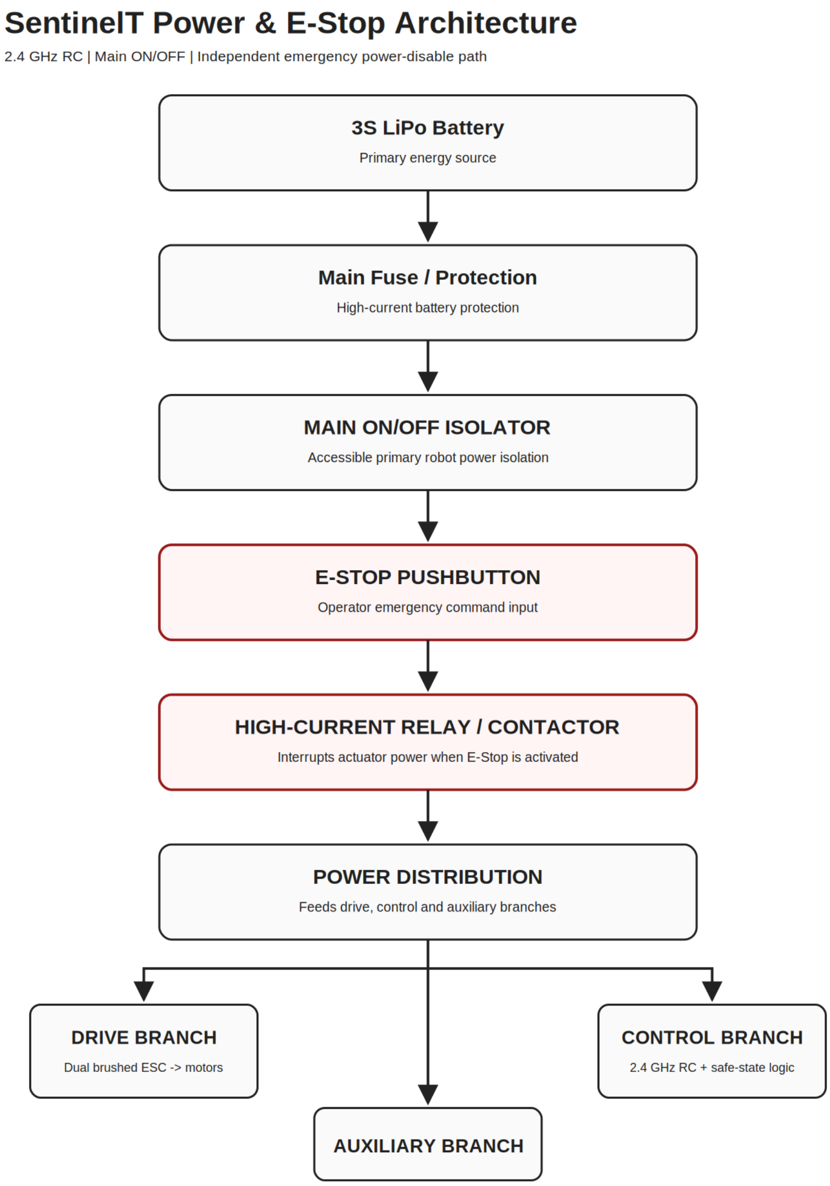
Blender model with repeatable Python footprint verification



- Verified CAD envelope: 318.16 mm x 495.00 mm x 405.99 mm.
- 204 physical mesh objects audited.
- Frames 1-180 checked.
- 0 failed frames; PASS 180/180.
- Digital verification completed before physical fabrication.

Electronics, Control & Safety

2.4 GHz RC with hardware isolation and E-Stop integration



- 3S LiPo power system and brushed drive controller.
- FlySky FS-i6X + X6B 2.4 GHz human-operated RC.
- Accessible main ON/OFF isolator.
- Mushroom E-Stop controls a separate high-current relay/contactor.
- RC signal loss and invalid commands return software to SAFE_DISABLED.

Components, Mass & Cost

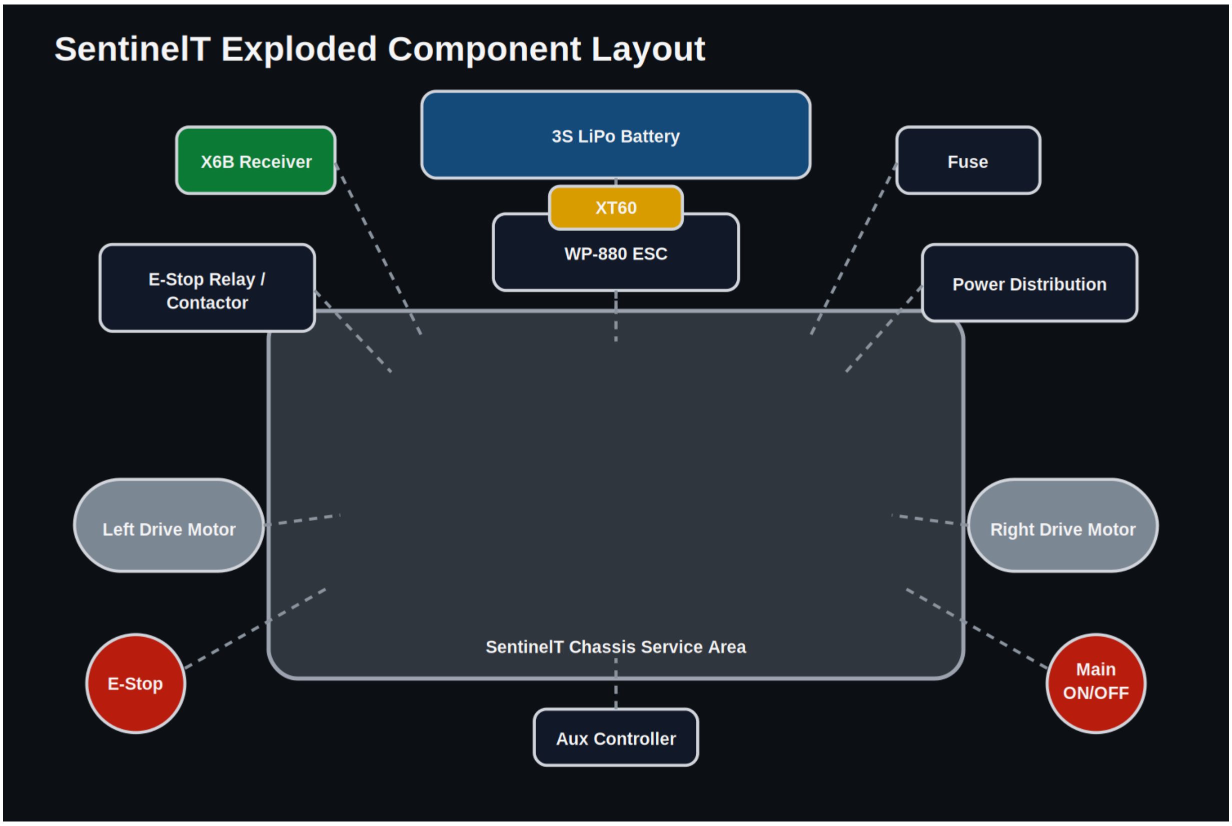
Current engineering implementation plan

4.650 kg

Engineering mass allocation

R 8,039.15

Current priced component subtotal



Final active-mechanism motor/controller and implementation total are verified after hardware matching.

Project Developer

TG Assegaai

- System concept and engineering architecture.
- Blender CAD modelling and mechanical packaging.
- Python footprint and mass-budget verification tools.
- 2.4 GHz RC control framework and safe-state logic.
- Main ON/OFF and E-Stop safety architecture.
- Component research, BOM development and technical documentation.

Signature: Thato Glen Assegaai